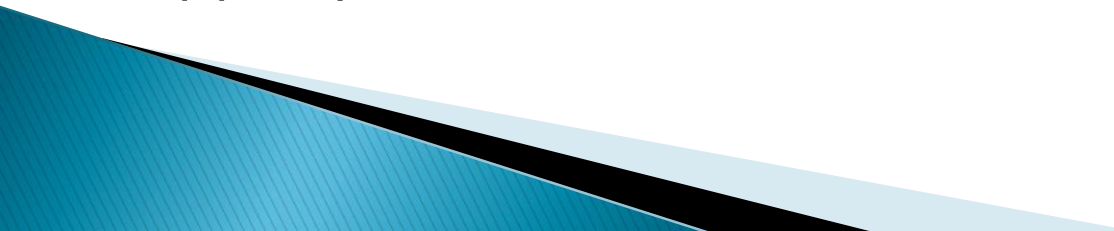


National Income:

- ▶ There are various concepts of national income. These are explained below one by one:
 - ▶ (1) Gross National Product (GNP).
 - ▶ (2) Net National Product (NNP)/National Income.
 - ▶ (3) Gross Domestic Product (GDP).
 - ▶ (4) National Income at Factor Cost.
 - ▶ (5) Personal Income.
 - ▶ (6) Disposable Personal Income.
- 

(1) Gross National Product (GNP):

▶ **Definition and Explanation of GNP:**

▶ The concept of gross national product (GNP) is comprehensive. It enables us to measure and analyze as to how much is the aggregate economic production of a country in a given period. The gross national product of a country (GNP) is defined as:

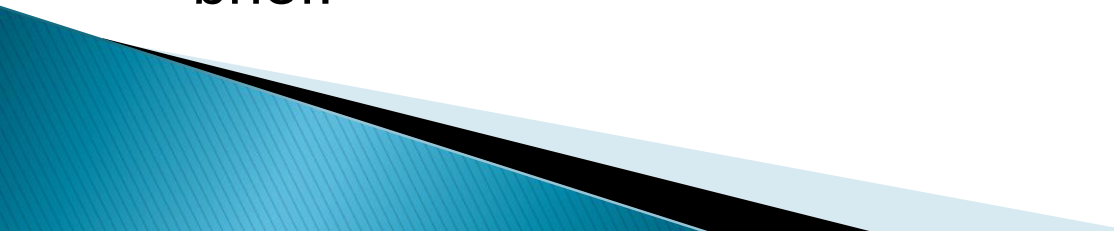
▶ "The total money value of all final goods and services produced by the residents of a country in one year period".

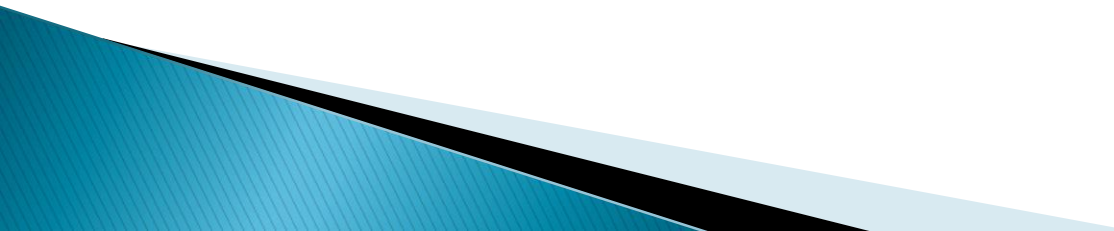
▶ In the words of W.C. Peterson:

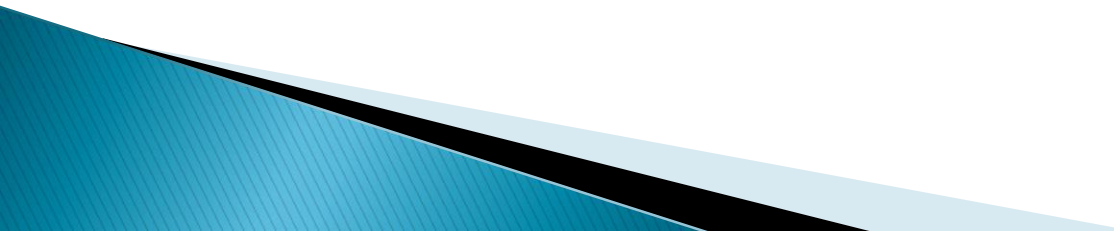
▶ "Gross National Product may be defined as the current market value of all final goods and services produced by the economy during an income period regardless of where the output is produced".



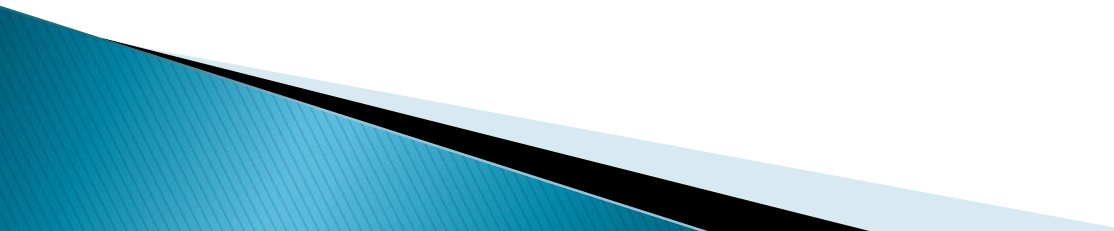
Components of Expenditure Approach in GNP:

- ▶ For measuring GNP at market price, the economists use **Expenditure Approach**. According to this approach:
 - ▶ There are four categories of expenditures which are added together to measure gross national product (GNP) at market price, (i) Consumption, (ii) Investment (iii) Government expenditure and (iv) Net exports.
 - ▶ These ***four types of expenditures*** are now explained in brief:
- 

- ▶ **i) Consumption Expenditure (C):** It includes all personal expenditure incurred by the citizens of a country on durable and non-durable goods in a period of one year.
 - ▶ **(ii) Investment (I):** It is the total expenditure incurred by firms or households on capital goods.
 - ▶ **(iii) Govt. expenditures (G):** It includes all types of expenditure incurred by Federal, Provincial, and Local Councils on the purchases of goods and services such as national defense, law and order, street lighting etc.
- 

- ▶ **(iv) Net Exports (X - M):** Net exports of goods and services are value of exports minus the value of imports.
 - ▶ **Formula for Gross Profit:**
 - ▶ **$GNP = C + I + G + (X - M)$**
 - ▶ Where:
 - ▶ C = consumption, I = investment, G = Govt. expenditure and X - M = Net exports
- 

(2) Net National Product (NNP)/National Income:

- ▶ **Definition and Explanation of NNP:**
 - ▶ ***"Net national Product or national income"*** at market prices is the net market money value of all the final goods and services produced in a country during a year.
 - ▶ It is found out by subtracting the amount of depreciation of the existing capital in a year from the market value of all final goods and services". If we deduct depreciation allowance from gross national product, we get Net National Product at current market price.
 - ▶ **Formula for Net National Product/National Income:**
 - ▶ **NNP at Market Price = GNP at Market Price - Depreciation**
- 

Example of NNP:

- ▶ Suppose, a person buys a machinery for manufacturing cloth for \$10000 only. He expects that this machinery will last ten years and after that period, it will be partially or completely worn out. He sets aside \$1000 every year from the gross national income as a depreciation reserve of the capital equipment.
- ▶ After the expiry of ten years, he accumulates \$10000 and with that money he replaces the old capital equipment which has lived its useful life and maintains capital intact. The sum of money, i.e., \$1000 which he annually deducts from the gross annual income, is known as ***depreciation allowance***.

3) Gross Domestic Product (GDP):

- ▶ It is a key concept in the national income. "**Gross domestic product (GDP)** is the total market value at current prices of all final goods and services produced within a year by the factors of production located within a country".
- ▶ If we add up the money value of all the final goods produced both by domestic and foreign owned factors annually in the country and valued at market prices, it will be called gross domestic product (GDP).
- ▶ According to **Shapiro**:
- ▶ "GDP is defined as a flow variable, measuring the quantity of final good and services produced" during a year".

Problems in Measuring GDP:

- ▶ The main problems of GDP are as under:
- ▶ **(i) Stress on final output.**
- ▶ While calculating the gross domestic product (GDP), the value of only those goods are added which have reached their final stage of production and are available for consumption. The primary or intermediate goods are not counted in GDP. For example, table made of wood is the final product. The wood used in making the table is a primary good. While calculating GDP, if we include the value of wood as a separate item and the value of table separate, it will be a case of double counting and this leads to inflated rise in GDP.

- ▶ **ii) Value added method.**

- ▶ Another way to avoid pitfall of double or multiple counting is to calculate only the added value of a particular commodity at its every stage of production. The result in both the cases will be the same.

- ▶ From the following example, the reader can easily understand as to how the danger of double or multiple counting can be avoided.

Stage of Production	Form of the Product	Price at Each Stage (\$)	Value Added at Each Process (\$)
1 st	Jungle Wood	0.25	0.25
2 nd	The price of wood after transporting to the city	0.38	0.13
3 rd	Paper manufacturing	2.00	1.62
4 th	Printing of book	5.00	3.00
5 th	Binding and title, etc.	6.00	1.00
6 th	Sale price	10.00	4.00
		\$23.63	\$10.00

- ▶ From the above example, it is clear that if we add up the value of the product at every stage of production, the total value of the book comes to \$23.63, while in fact it is priced at \$10 only.
 - ▶ So we come to the conclusion that while adding the value of the book to the gross national product, we should either include the final price of the book which is \$10 or we should add up the added value at each stage in the process of production. But we are not to count the value of a particular commodity more than once. If we do so, the gross product will be overestimated. The computation of GDP by this method is not popular.
- 

- ▶ (iii) **Non-Productive transactions are excluded from GDP.**
- ▶ In order to measure the economic well-being of a society in a year, the non-productive transactions are excluded from the Gross Domestic Product. There are ***one major type of non-productive transactions***, namely: (a) Purely financial transactions
- ▶ Under purely financial transaction (i) all public transfer payments which do not add to the current flow of goods **such as social security payments, relief payments** and
- ▶ (ii) all private financial transactions such as **receipt of money by a student from his father** which make no contribution in current production are all excluded from GDP.

▶ **(iv) Other transactions.**

▶ There are a few other transactions which are not included in GDP. For example, persons working in their own houses without any payment through the market. For example, a house wife takes care of house and children. Since she is not paid, therefore, the value added by her is not included in GDP.

▶ **(v) Exclusion of output production abroad.**

▶ GDP is the value of output produced by factors of production located within a country. It excludes the output produced abroad by domestically owned factors of production.

Distinction between GDP and GNP:

- ▶ Gross domestic product is the total market value of all final goods and services produced by factors of production within a nation's border during a period of one year. In other words GDP is a flow of production produced within the country by domestically located resources in a year.
- ▶ Gross national product (GNP) on the other hand, is the measure of all final goods and services produced by the citizens within their own country as well as outside the country during a period of one year. In other words, GNP expresses the money value of flow of goods and services produced within the country and the net income received from abroad during a period of one year. Thus when we move from GDP to GNP, we add factor income receipts from foreigners and subtract factor income payments to foreigners.
- ▶ **Formula for GDP:**
- ▶ **$GDP = GNP - \text{Net Foreign Income from Abroad}$**

(4) National Income

Definition and Explanation:

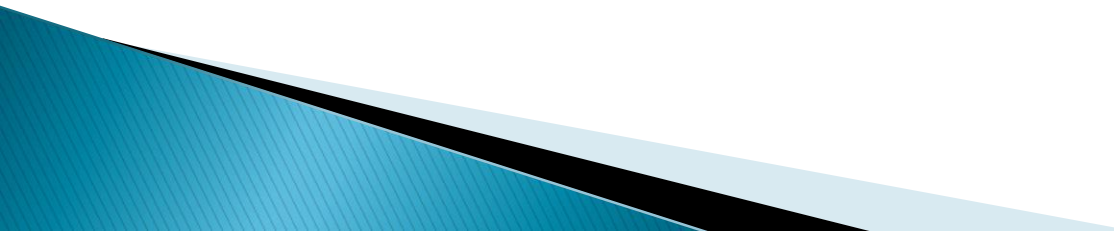
- ▶ National income can be estimated in terms of either output or total income. When national income is measured by adding together all income payments made to the factors of production in a year, it is called national income
- ▶ . In the words of **J. Sloman**:
- ▶ ***"National income (NI) or national income at factor cost*** is the aggregate earning of the four factors of production (land, labor, capital and organization) which arise from the current production of goods and services by the nations' economy".

Components of National Income

- ▶ The main components of national income at factor cost are as follows:
 - ▶ **(i) Compensation to employees:** It is the largest component of national income. It consists of wages and salaries paid by the firms to the workers for their labor services.
 - ▶ **(ii) Interest:** Interest is the payment for the use of funds in a year. The payment is made by private businesses to households who have lent money to them.
 - ▶ **(iii) Rent:** Rent is all income earned by individuals for the use of their real assets such as building, farms etc.
 - ▶ **(iv) Profit:** Profit is the amount which is left after compensation to employees, rent, and interest has been paid out. The sum of compensation to employees, interest, rent and profit is supposed to equal national income at factor cost.
- 

(5) Personal Income:

▶ **Definition and Explanation:**

- ▶ National income is the sum of factor income. In other words, it is the income which individuals receive for doing productive work in the form of wages, rent, interest and profits.
 - ▶ Personal income, on the other hand, includes all income which is actually received by all individuals in a year. It includes income which is not directly earned but is received by individuals.
 - ▶ **For example**, social security payments, welfare payments are received by households but these are not elements of national income because they are transfer payments.
- 

▶ In the same way, in national income accounting, individuals are attributed income which they do not actually receive. For example, undistributed profits, employee's contribution for social security corporate income taxes etc. are elements of national income but are not received by individuals. Hence they are to be deducted from national income to estimate the personal income.

▶ **Formula for Personal Income:**

▶ **$PI = NI + \text{Transfer Payments} - \text{Corporate retained earnings, income taxes, social security taxes}$**



(6) Disposable Personal Income:

- ▶ **Definition and Explanation:**

- ▶ Disposable personal income is the amount which is actually at the disposal of households to spend as they like. It is the amount which is left with the households after paying personal taxes such as income tax, property tax, national insurance contributions etc.

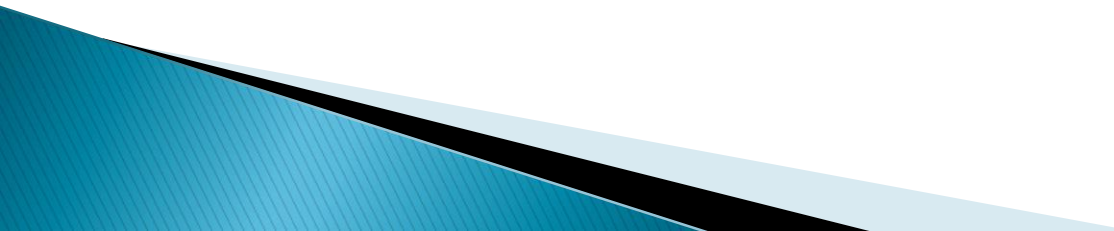
- ▶ **Formula for Disposable Personal Income:**

Disposable personal income = Personal Income - Personal Taxes

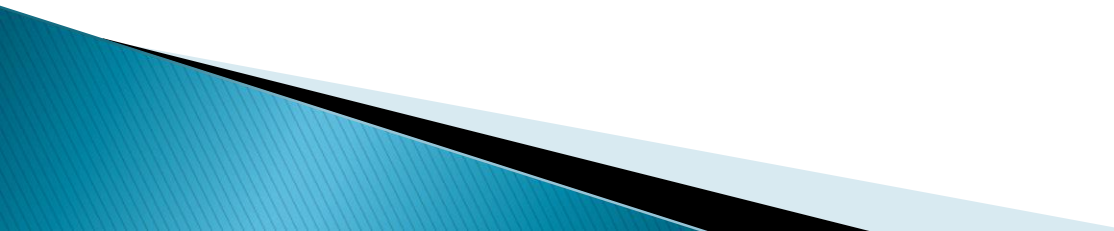
- ▶ The concept of disposable personal income is very important for studying the consumption and saving behavior of the individuals. It is the amount which households can spend and save.

- ▶ **Disposable personal Income = Consumption + Saving**

- ▶ **DPI = C + S**



Methods of Computing/Measuring National Income:

- ▶ There are three ***methods of measuring national income*** of a country. They yield the same result. These methods are:
 - ▶ (1) The Product Method.
 - ▶ (2) The Income Method.
 - ▶ (3) The Expenditure Method.
- 

(1) Product Method or Value Added Method:

▶ **Definition and Explanation:**

▶ Goods and services are counted in **gross domestic product (GDP)** at their market values. The product approach defines a nation's gross product as that market value of goods and services currently produced within a nation during a one year period of time.

▶ The ***product approach*** measuring national income involves adding up the value of all the final goods and services produced in the country during the year. Here we focus on various sectors of the economy and add up all their production during the year. The main sectors whose production value is added up are:

- ▶ (i) agriculture (ii) manufacturing (iii) construction (iv) transport and communication (v) banking (vi) administration and defense and (vii) distribution of income

Precautions for Product Method or Value Added Method:

- ▶ There are certain precautions which are to be taken to avoid miscalculation of national income using this method. These in brief are:
 - ▶ **(i) Problem of double counting:** When we add up the value of output of various sectors, we should be careful to avoid double counting. This pitfall can be avoided by either counting (the final value of the output or by including the extra value that each firm adds to an item.
 - ▶ **(ii) Value addition in particular year:** While calculating national income, the values of goods added in the particular year in question are added up. The values which had previously been added to the stocks of raw material and goods have to be ignored. GDP thus includes only those goods, and services that are newly produced within the current period

▶ **(iii) Stock appreciation:**

▶ Stock appreciation, if any, must be deducted from value added. This is necessary as there is no real increase in output.

▶ **(iv) Production for self-consumption:**

▶ The production of goods for self-consumption should be counted while measuring national income. In this method, the production of goods for self-consumption should be valued at the prevailing market prices.



(2) Expenditure Method:

► Definition and Explanation:

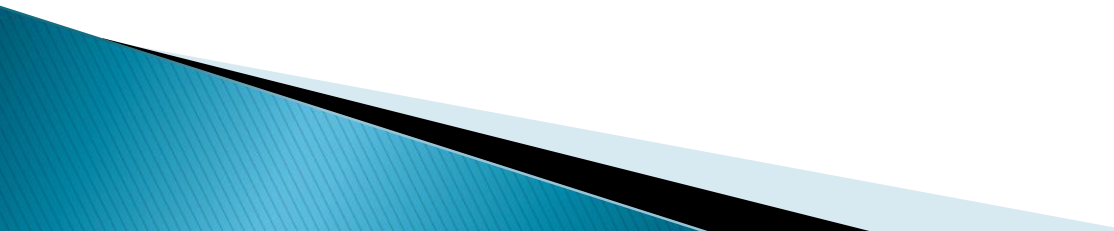
► ***The expenditure approach*** measures national income as total spending on final goods and services produced within nation during a year. The expenditure approach to measuring national income is to add up all expenditures made for final goods and services at current market prices by households, firms and government during a year. Total aggregate final expenditure on final output thus is the sum of four broad categories of expenditures:

► (i) Consumption (ii) investment (iii) government expenditure and (iv) net export.

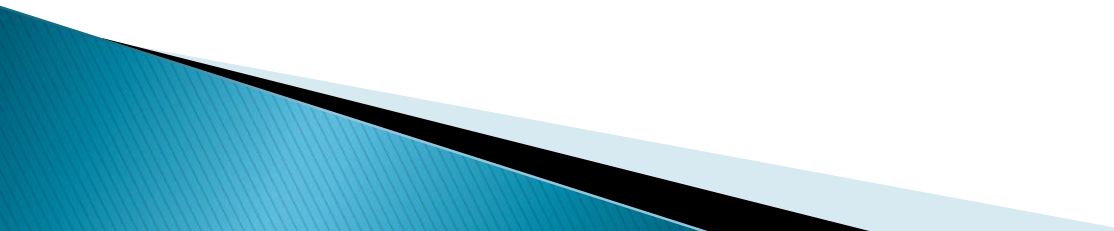
► **(i) Consumption expenditure (C):** Consumption expenditure is the largest component of national income. It includes expenditure on all goods and services produced and sold to the final consumer during the year.

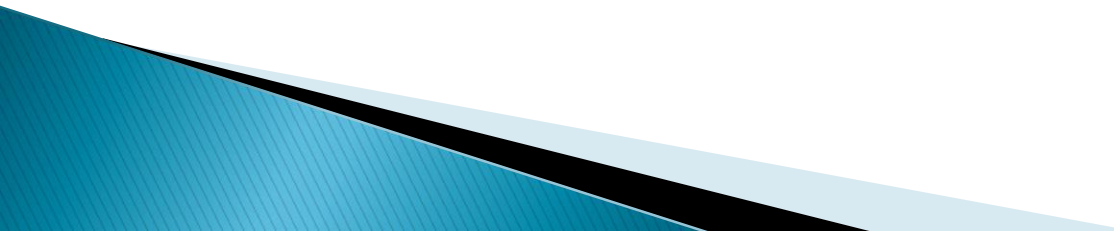
- ▶ **ii) Investment expenditure (I):** Investment is the use of today's resources to expand tomorrow's production or consumption. Investment expenditure is expenditure incurred on by business firms on (a) new plants, (b) adding to the stock of inventories and (c) on newly constructed houses.
- ▶ **(iii) Government expenditure (G):** It is the second largest component of national income. It includes all government expenditure on currently produced goods and services but excludes transfer payments while computing national income.
- ▶ **(iv) Net exports (X - M):** Net exports are defined as total exports minus total imports.
- ▶ National income calculated from the expenditure side is the sum of final consumption expenditure, expenditure by business on plants, government spending and net exports.
- ▶ **$NI = C + I + G + (X - M)$**

Precautions for Expenditure Method:

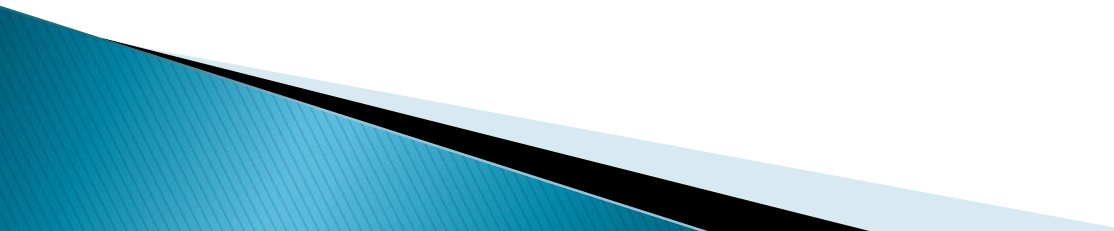
- ▶ While estimating national income through expenditure method, the following precautions should be taken:
 - ▶ (i) The expenditure on second hand goods should not be included as they do not contribute to the current year's production of goods.
 - ▶ (ii) Similarly, expenditure on purchase of old shares and bonds is not included as these also do not represent expenditure on currently produced goods and services.
 - ▶ (iii) Expenditure on transfer payments by government such as unemployment benefit, old age pensions, interest on public debt should also not be included because no productive service is rendered in exchange by recipients of these payments.
- 

(3) Income Approach:

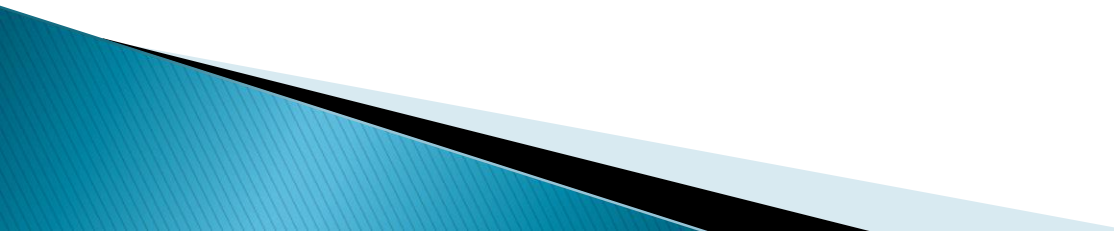
- ▶ ***Income approach*** is another alternative way of computing national income. This method seeks to measure national income at the phase of distribution. In the production process of an economy, the factors of production are engaged by the enterprises.
 - ▶ They are paid money incomes for their participation in the production. The payments received by the factors and paid by the enterprises are wages, rent, interest and profit.
 - ▶ National income thus may be defined as the sum of wages, rent, interest and profit received or occurred to the factors of production in lieu of their services in the production of goods.
 - ▶ Briefly, national income is the sum of all income, wages, rents, interest and profit paid to the four factors of production. The four categories of payments are briefly described below:
- 

- ▶ **(i) Wages:** It is the largest component of national income. It consists of wages and salaries along with fringe benefits and unemployment insurance.
 - ▶ **(ii) Rents:** Rents are the income from property received by households.
 - ▶ **(iii) Interest:** Interest is the income private businesses pay to households who have lent the business money.
 - ▶ **(iv) Profits:** Profits are normally divided into two categories (a) profits of incorporated businesses and (b) profits of unincorporated businesses (sole proprietorship, partnerships and producers cooperatives).
- 

Precautions for Income Approach:

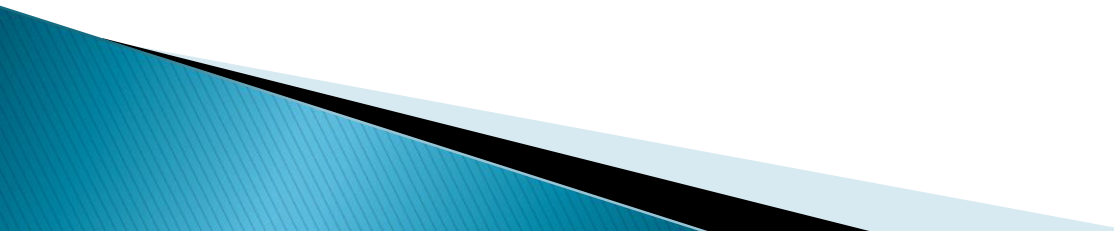
- ▶ (i) Transfer payments such as gifts, donations, scholarships, indirect taxes should not be included in the estimation of national income.
 - ▶ (ii) Illegal money earned through smuggling and gambling should not be included.
 - ▶ (iii) Windfall gains such as prizes won, lotteries etc. is not be included in the estimation of national income.
 - ▶ (iv) Receipts from the sale of financial assets such as shares, bonds should not be included in measuring national income as they are not related to generation of income in the current year production of goods.
- 

Why Three Methods of Computing/Measuring National Income are Equal:

- ▶ The three approaches used for measuring national income give the same result. The reason is the market value of goods and services produced in a given period by definition are equal to the amount that buyers must spend to purchase them. So the product approach which measures market value of goods and services produced and
 - ▶ The expenditure approach which measures spending should give the same measure of economic activity.
 - ▶ Now as regards the income approach, the seller's receipts must equal what the buyers spend. The seller's receipts in turn equal the total income generated by the economic activity. Thus, total expenditure must equal total income generated implying that the expenditure and income approach must also produce the same result.
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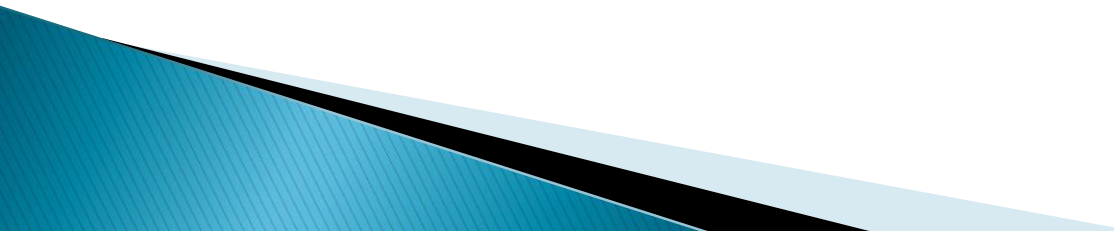
Circular Flow of National Income in a Two Sector Economy or Circular Flow Model:

▶ Definition of Circular Flow Model:

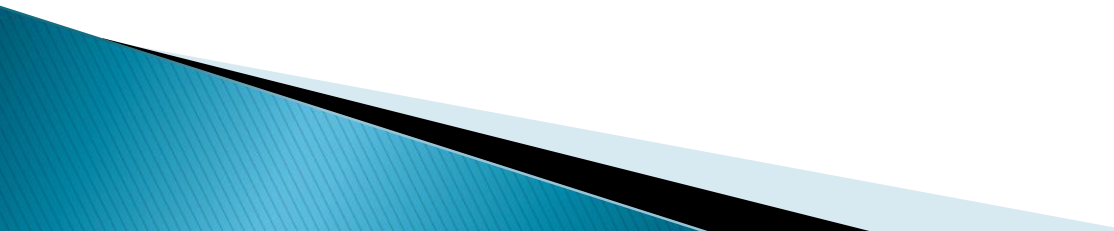
- ▶ A simple ***circular flow model*** of the macro economics containing two sectors (business and household) and two markets (product and factor) that illustrates the continuous movement of the payments for goods and services between producers and consumers. The payment flow between the two sectors and two markets is conveniently divided into four segments representing consumption expenditures, gross domestic product, factor payments, and national income.
 - ▶ The modern economy is a monetary economy. In the modern economy, money is used as a medium of exchange. While analyzing the ***circular flow of income in a two sector model of the economy***, we assume:
- 

▶ **Assumptions of Circular Flow Model:**

- ▶ (i) There are only two sectors in the economy, household sector and business sector.
- ▶ (ii) The business sector (or the firms) hires factors of production owned by the household sector and it is the sole producer of goods and services in the economy.
- ▶ (iii) The household sector (or the households) is the sole buyer of goods and services. It spends its entire income on the goods and services produced by the business sector. They are also suppliers of labor and various of other factors of production.
- ▶ (iv) The business sector sells the entire output to households. It does not store. There are, therefore, no inventories.

- ▶ (v) There are no savings and investment in the economy.
 - ▶ (vi) The household sector receives income by selling or renting the factors of production owned by it.
 - ▶ (vii) Government does, not exist for all such practical purposes (No public expenditures, no taxes, no subsidies, no social insurance contribution, etc.).
 - ▶ (viii) The economy is closed one having no international trade relations.
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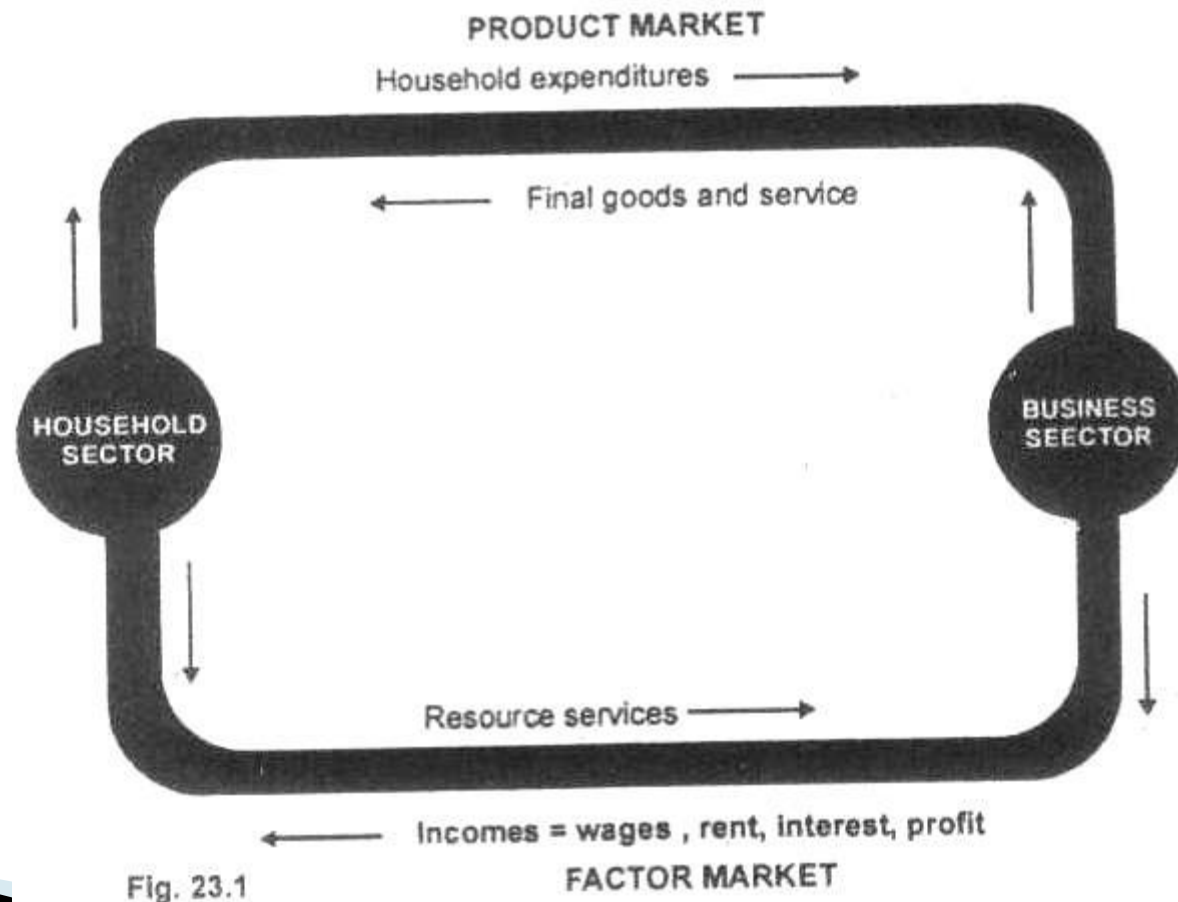
▶ Principles of Circular Flow of National Income:

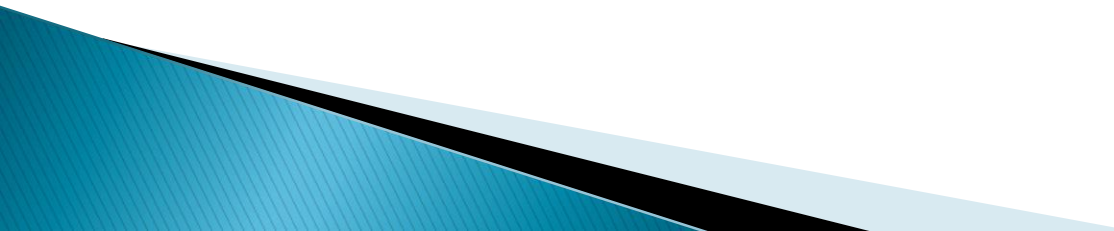
- ▶ In the simple circular flow of income and product, there are two principles which are involved.
 - ▶ **First.** In the business transactions, the sellers of goods receive exactly the same amount which the buyers spend on them.
 - ▶ **Second.** The goods and services flow in one direction and money payment flow in the other direction.
- 

► **Explanation of Circular Flow of National Income:**

- In a two sector economy, there are business firms which produce goods and services.
- The other sector is households which supplies their factors services to the firms and also buy goods and services produced by them.
- The households supply the economic resources to the firms and receive payments in terms of money. There is, thus, a flow of money corresponding to the flow of economic resources. These money incomes are spent by households on goods and services produced by the firms. With this the money comes back to the firms. This circular flow of income in fact is the mutual dependence of the two sectors of modern economy.

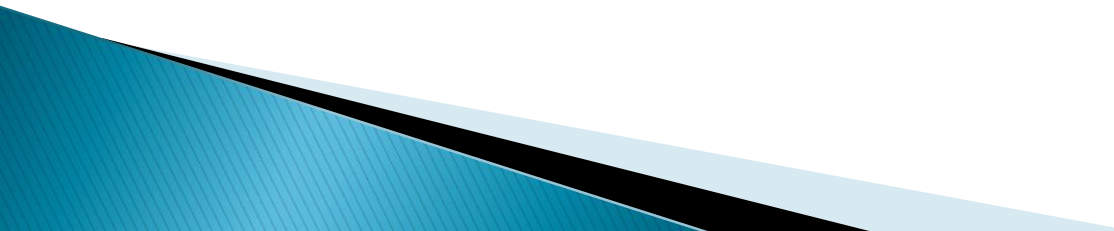
- ▶ **Diagram of Circular Flow of Income:**
- ▶ The circular flow of income in a two sector economy is explained with the help of **figure 23.1**.



- ▶ In this figure, it is shown that the economy consists of two sectors (1) households and business. In the upper top of this figure, the resources such as land, capital, labor and entrepreneurial ability flow from households to business firms as indicated by the arrow mark. In opposite direction to this, money flows from business firms to the households as factors payments such as rent, wages, interest and profit.
 - ▶ In the lower pipe line, money flows from households to firms as consumption expenditure made by the households on the goods and services produced by the firms. The flow of goods and services is in opposite direction from business firms to households. We, thus, find that money flows from business firms to households as factor payments and then it flows back from households to firms. Thus there is in fact a circular flow of income. This circular flow of money or income continues year after year. This is how the economy functions.
- 

Difficulties/Problems in the Measurement of National Income:

- ▶ According to **Kuznets**, the **measurement of national income** is a complicated problem and is best with the following difficulties:
- ▶ **(i) Non-availability of statistical material:** Some persons like electricians, plumbers, etc., do some job in their spare time and receive income. The state finds it very difficult to know the exact amount received from such services. This income which, should have been added to the national income is not recorded due to be lack of full information of statistics material.
- ▶ **(ii) The danger of double counting:** While computing the national income, there is always the danger of double or multiple counting. If care is not taken in estimating the income, the cost of the commodity is likely to be counted twice or thrice and national income will be overestimated.

- ▶ **(iii) Non-marketed services:** In estimating the national income, only those services are included for which the payment is made. The unpaid services, or non-marketed services are excluded from the national income.
 - ▶ **(iv) Difficulty in assessing the depreciation allowance:** The deduction of depreciation allowances, accidental damages, repair, and replacement charges from the national income is not an easy task. It requires high degree of judgment to assess the depreciation allowance and other charges.
 - ▶ **(v) Housing:** A person lives in a rented house. He pays \$5000 per month to the landlord. The income of the landlord is recorded in the national income. Let us suppose that the tenant purchases the same house from the landlord. Now the income of the owner occupant has increased by \$5000. Is it not justifiable to include this income in the national income? Should or should not this income be recorded in the national income is still a controversial question.
- 

- ▶ **(vi) Transfer earnings:** While measuring the national income, it should be seen that transfer payments should not become a part of national income. The payments made as relief allowance, pensions, etc. do not contribute towards current production. So they should be excluded from national income.
 - ▶ **(vii) Self-consumed production:** In developing countries, a significant part of the output is not exchanged for money in the market. It is either consumed directly by producers or bartered for other goods. This unorganized and non-monetized sector makes calculation of national income difficult.
 - ▶ **(viii) Price level changes:** National income is measured in money terms. The measuring rod of money itself does not remain stable. This means that national income can change without any change in output.
- 